

Ali Abdari

Computer Vision and Machine Learning Engineer

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🌐 Website

🐙 GitHub

🌐 LinkedIn



Date of Birth: March 26, 1994 - Tel:(+39)3792455496

Profile

As an applied researcher, I have been actively employing advanced Artificial Intelligence and Machine Learning techniques to address a wide range of problems involving diverse data types, including tabular, textual, visual, and 3D data. In particular, my research has focused on both single-modal and cross-modal data analysis, with my PhD work centered on retrieving and matching complex 3D scenarios with textual descriptions. I am now seeking industry opportunities where I can apply my skills and expertise as a **Computer Vision and Machine Learning Engineer** to solve real-world challenges.

Commercial and Academic Experiences

- 2023 – 2024 **Teaching Assistant** of Deep Learning, University of Udine.
- 2021 – 2023 **Computer Vision and Machine Learning Researcher** at Kharazmi University. (Action Recognition and Violence Detection in Video, Compressed Video Analysis)
- 2022 **Computer Vision Engineer** at Caica Company. (Primates Behavioural Analysis via Video, Fatigue Detection System using Face Key Points, AI-based Assistant Systems for patients)
- 2019 – 2021 **Computer Vision Engineer** at Iran's National Elites Foundation. (Surveillance Video Synopsis, Traffic Video Analysis, Surveillance Videos Event Detection)
- 2018 – 2020 **Android Developer** at Mosbat Co. (Designing and Developing Android applications, Maintaining and adding features to current projects)
- 2017 – 2018 **Python Developer** at Vestaak Co. (Developing a Recommender System for Academic Purposes)
- 2017 – 2019 **Teaching Assistant** of Image Processing and Neural Networks, Kharazmi University.
- 2013 – 2016 **Teaching Assistant** of Data structure, Automata theory, Design and Analysis of Algorithms, Introduction to Programming and Advanced Programming, Kharazmi University.

Skills

- Languages **English** (C1, 7.5 IELTS band score), **Italian** (B2), **Persian** (Native), **Arabic** (B1).
- Coding **Python**, **C++**, **C**, **MATLAB**, **R**, **Java**, **PHP**, **C Sharp**, **SQL**, **Julia**.
- Databases **MySQL**, **SQL Server**, **SQLite**, **MongoDB**.
- Web Dev **HTML**, **css**, **JavaScript**, **Bootstrap**.
- ML/DL Frameworks **PyTorch**, **Keras**, **HuggingFace**, **OpenCV**, **Dlib**, **MediaPipe**, **NumPy**, **Pandas**, **Spacy** & **NLTK**, **Matplotlib**, **SciKit-Learn**, and **SHAP**.

Selected Projects

- Cross Modal Processing **3D scene/Point Cloud Text Alignment Retrieval**, **Contrastive Learning**
- Video Processing **Action Recognition in raw and compressed video**, **Primates Behaviour Analysis**, **Traffic Video Analysis**, **Video Synopsis**, **Multi-Object Tracking in surveillance videos**, **Fatigue Detection using eye features**

Selected Projects (continued)

Image Processing	■	OCR, Water Marking, Segmentation, Classification
Tabular data	■	Prediction of Agricultural and Meteorological data, Genetic data processing
Miscellaneous	■	EEG Signal Analysis, Risk Assessment in Autonomous Vehicles, Vehicle Position Prediction using AI, Web Crawling to Collect Data

Academic Activities and Achievements

- 2025
 - Organizing 4th edition of CV4Metaverse Workshop at CVPR 2025, USA.
 - Selected as an Organizer Chair of the IRCDL 2025 conference, Italy.
 - Supervising a Ph.D. student in agricultural and meteorological data processing using ML.
 - Supervising a researcher in plant genetic data processing using ML.
- 2024
 - Organizing 3rd edition of CV4Metaverse Workshop at ECCV 2024, Italy.
 - Organizing Metaverse Apartment Retrieval Challenge at CV4Metaverse at ECCV 2024, Italy.
 - Selected in the top 15% in Summer School on Computer Vision in the Age of LLMS, ICVSS, Italy.
- 2023
 - Selected in the top 25% in Summer School on Large-Scale AI, ELLIS, Modena, Italy
- 2022
 - Granted €65K Ph.D. scholarship at University of Naples Federico II, Italy.
- 2016
 - Admitted to Master's program among top 10% without entrance exam.

Education

- 2023 – Dec 2025
 - **Ph.D. Student, University of Naples Federico II, University of Udine, Italy** Artificial Intelligence.
Area: *Complex 3D Scene and Metaverse Retrieval*
Supervisor: Prof. Giuseppe Serra
- 2016 – 2019
 - **M.Sc. Kharazmi University, Iran** Computer Science.
Thesis title: *Action Recognition in Compressed Video*.
Supervisor: Prof. Azadeh Mansouri - Grade: 18.34/20
- 2012 – 2016
 - **B.Sc. Kharazmi University, Iran** Computer Engineering.
Thesis title: *Reporting Financial affairs using Business Intelligence*.
Supervisor: Prof. Zahra Nilfroushan - Grade: 16.98/20

Selected Publications ([Google Scholar](#))

- 1 A. Falcon, **A. Abdari**, and G. Serra, "Alcer3d: Adaptive learning constraints for enhanced retrieval of complex indoor 3d scenarios," *IEEE Transactions on Multimedia*, 2025.
- 2 **A. Abdari**, A. Falcon, and G. Serra, "A language-based solution to enable metaverse retrieval," in *International Conference on Multimedia Modeling*, Springer, 2024, pp. 477–488.
- 3 **A. Abdari**, A. Falcon, and G. Serra, "Adoctera: Adaptive optimization constraints for improved text-guided retrieval of apartments," in *Proceedings of the 2024 International Conference on Multimedia Retrieval*, 2024, pp. 1043–1050.
- 4 N. Honarjoo, **A. Abdari**, and A. Mansouri, "Violence detection in compressed video," *Multimedia Tools and Applications*, pp. 1–14, 2024.
- 5 **A. Abdari**, A. Falcon, and G. Serra, "Farmare: A furniture-aware multi-task methodology for recommending apartments based on the user interests," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 4293–4303.